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JavaScript

For beginners

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The basics of learning JavaScript

what you should know after learning

- understand what JavaScript is and its uses
- Be able to write JS code
- Use variables, data types, and operators
- Write simple functions and control flow statements
- Run JS in a browser console or simple webpage

What is JavaScript?

- JavaScript is a Programming language that makes websites interactive
- it allows you to add dynamic content, respond to user actions, and create applications

Why learn JavaScript?

- One of the most popular programming languages
- Used in almost every website
- Enables you to create websites, apps. And games

How JavaScript works

١. Runs in the browser or on a server (Node.js)

٢. Can be written directly inside HTML using <**Script**> tags

٣. The browser reads and executes code line by line

Your first code example

```
<Script>
    console.log("Hello, World!")
</Script>
```

- Prints “Hello, World!” in the console

Tips for beginners

- Start with small programs
- Experiment with different values
- Don’t be afraid of errors they’ll help you learning
- Practice everyday

unit one

(JS Intro & Basics)

١. Introduction

٢. Where to use JS

٣. Displaying output

٤. Syntax rules

٥. Statements

٦. Comments

(Variables & Operators)

١. Variables - Var/Let/Const

٢. Operators – Arithmetic, assignment, comparison

(Functions & Objects)

١. Functions – Defending & calling functions

٢. Objects – JS objects & Properties

٣. Events – Handling events

(String & Numbers)

١. Strings - JS strings
٢. Templates – String templates
٣. Numbers – JS numbers, Math, Random

(Arrays & Dates)

١. Arrays JS arrays
٢. Dates JS date objects

(Logic & ٤Loops)

١. Booleans - True/False values
٢. Comparisons – Comparison operators
٣. If/Else – Conditional statements
٤. Switch – Switch case
٥. Loops – Fr, While, Do-while
٦. Break – Loop control

(Errors & Modules)

١. Errors – Error handling
٢. Modules – JS modules

NOTE : we’re working with visual code, you can use any website and right click on inspect element.

JavaScript intro & basics

Why we should learn JavaScript?

JavaScript is one of the ٣ languages every web developer should know
(HTML, CSS, JavaScript)
١- HTML : Hyper Text Markup Language.
Its the standard language used to create and structure content on the web.

٢- CSS : Cascading Style Sheets
It’s the language used to style and design web pages that are built with HTML.

I**.e : HTML = the skeleton (Structure),**
CSS = the skin, clothes, and colors (style & layout).
JavaScript = the brain (interactivity & logic)

J**avaScript :** can change how website behave look and interact with you

JavaScript can change HTML content by : getElementById()

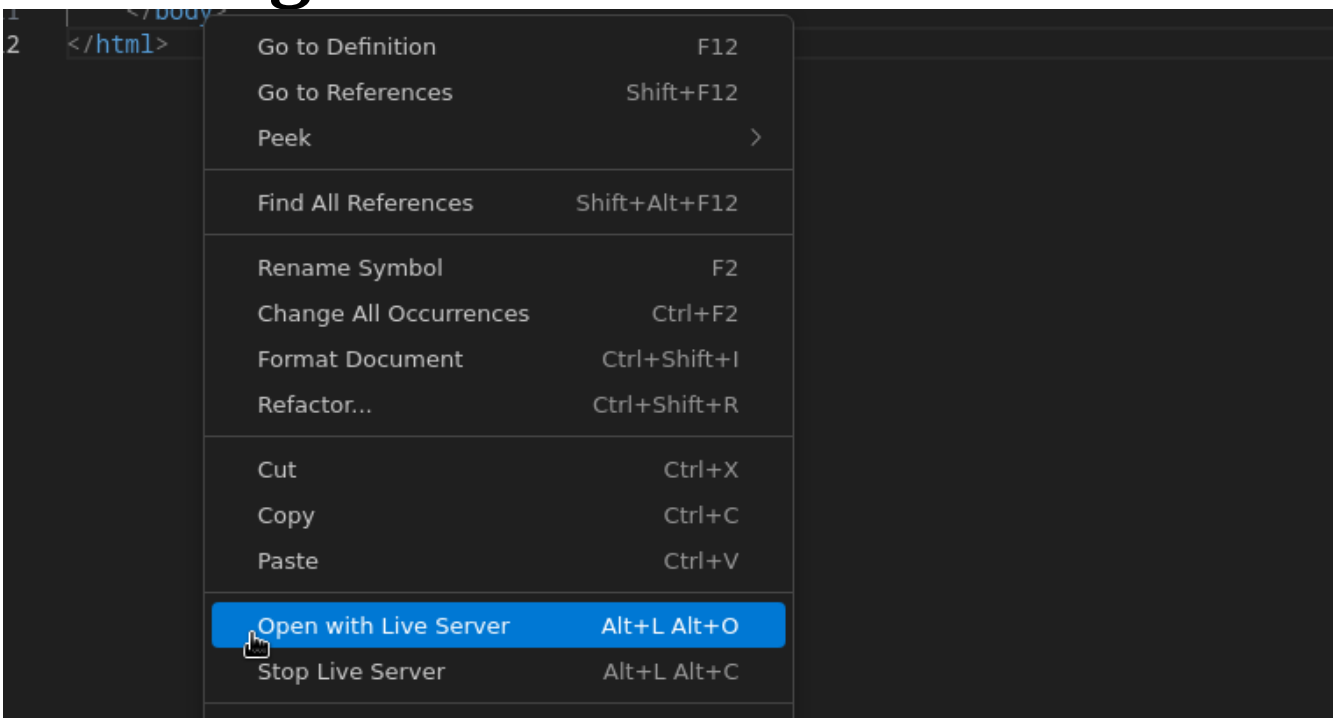
```
<html>
    <body>
<h2>What can JavaSript do?</h2>
<p id = "demo">JavaScript can change HTML content</p>
<button type="button" onclick='document.getElementById("demo").innerHTML = "Hello
JavaScript!"'>Click Me!</button>
    </body>
</html>
```

document.getElementById ("demo").innerHTML = "Hello JavaScript";
it finds an HTML element on the page with the ID “demo” and then changes its content to display “Hello JavaScript”.

١. document = the page
٢. getElemenById(“demo”) = find element with ID “demo”
٣. innterHTML = the content inside that element
٤. (=) changes it
٥. “Hello JavaSCript” = the new text string

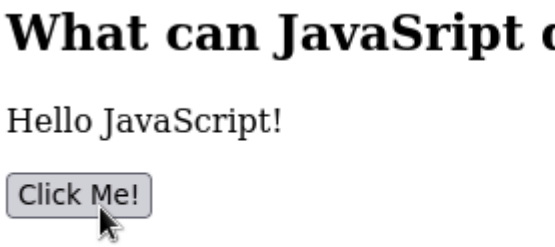
NOTE : JavaScript accepts both double and single quotes

how to get the results :



the results :

after clicking :



The lightbulb test

first you have to download ٢ images first “offlightbulb” and second “onlightbulb”

onlightbulb

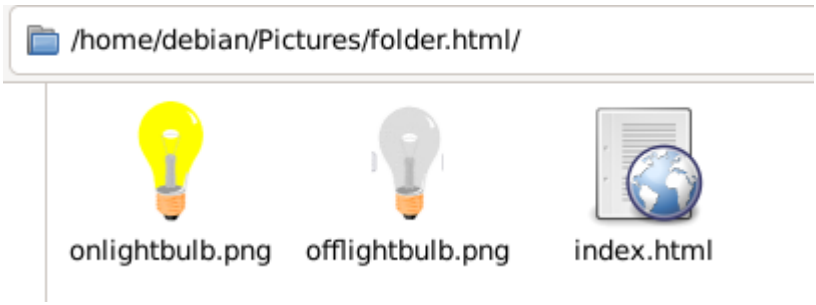
offlightbulb

so lets assume that the image format to be .png

٢. second step you have to make a folder

٣. add the two images to that folder

٤. create an index file and name it “index.html” in example

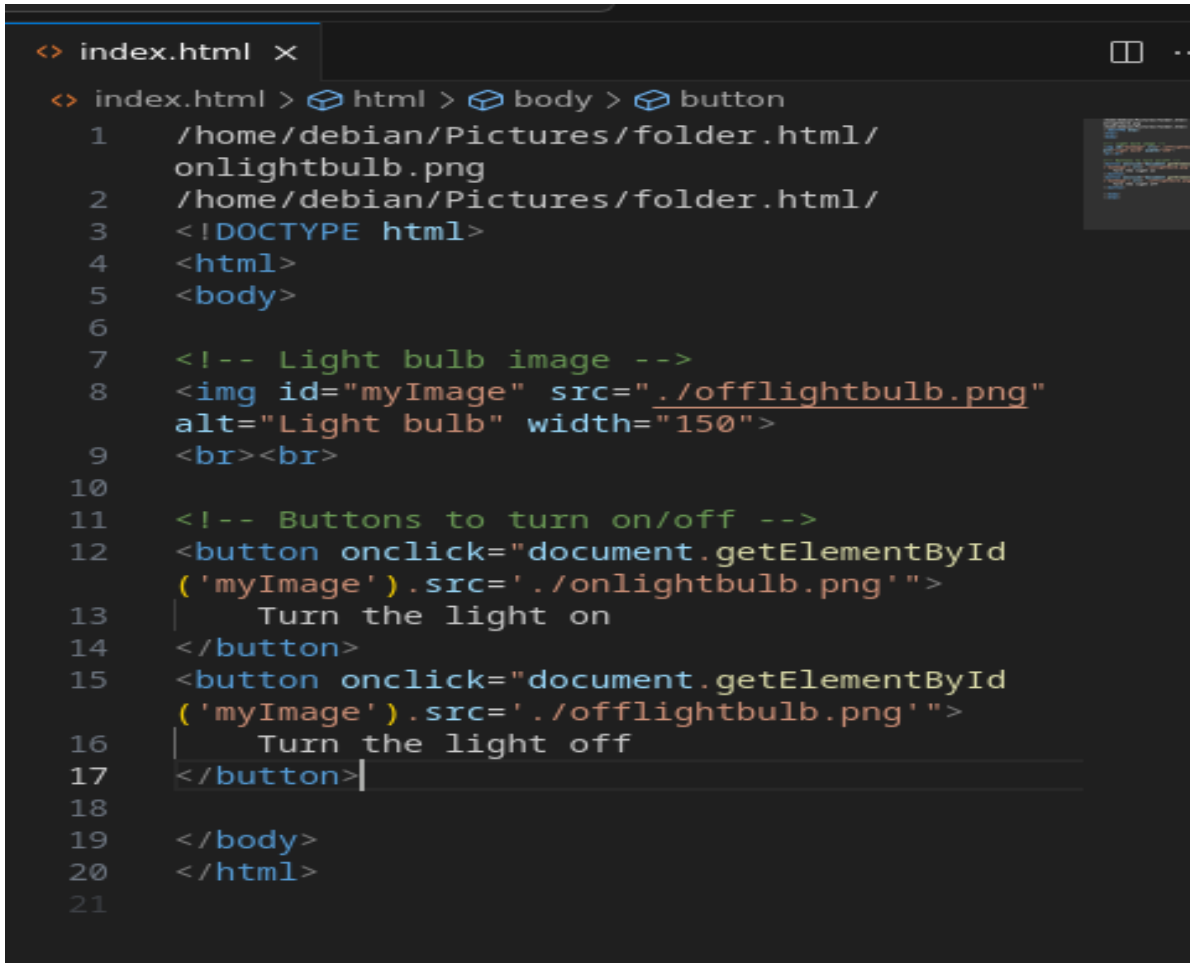


it should look like this on linux

NOTE: **make sure you've added these images to the index file**

Second step (visual code)

on visual code you should write this **code**



third step (live server)

What’s live server ?

live server : A VS code extension that launches a local web server and auto-refreshes your browser whenever you edit HTML/CSS/JS

how can I download live server on my visual code?

Steps to download live server :

١. open VS Code extensions (Ctrl+Shift+X)
٢. search “Live Server” Install (by Ritwick Dey)
٣. Open your HTML file – Right click – “Open with Live Server” OR click “Go Live” at the buttom-right
٤.Edit files – browser auto-refreshes

Final steps :

١. Open visual studio code
٢. Open your index.html file using visual code
٣. Add your code.
٣.Open the file with live server to see it live

Some code functions :

<!DOCTYPE html> : Declares the document type and version of HTML. (Document type).

<html>: Root element of the HTML document. All other HTML content

<head> : Contains meta-information about the page (not displayed

<meta charset="UTF-٨">: Specifies character encoding for the page.

<title>Light Bulb Example</title>: Sets the title of the page, shown in the browser tab.

<body>: Everything the user sees (text, images, buttons) goes here.

<h٢>What Can JavaScript Do?</h٢>: Provides a subheading; browsers usually display it larger and bold.

** JavaScript can change HTML attribute values. :**
function : Paragraph element with inline style
Explanation: Displays text. style="color:red;" makes the text red.

style → inline CSS for this element

color:red; → sets text color to red

<img id="myImage" src="./offlightbulb.png" alt="Light bulb" width="١٥٠"> :
function: it Displays an image.
Explanation:

id="myImage" → uniquely identifies this image for JS manipulation.

src="./offlightbulb.png" → file path of the image (./ = current folder).

alt="Light bulb" → text fallback if image doesn't load.

width="١٥٠" → sets image width to ١٥٠ pixels.

**

 :**
Function: Line break.
Explanation: Adds vertical spacing between elements. Two
s = two lines of space.

<button onclick=
"document.getElementById('myImage').src='./
onlightbulb.png'">Turn the light on</button>:

Function: Creates a clickable button that runs JavaScript when clicked.
Explanation:

document.getElementById('myImage') → selects the element with id="myImage"

.src='./onlightbulb.png' → changes the src attribute to show a different image

Inner text "Turn the light on" → the label displayed on the button

<button onclick=

"document.getElementById('myImage').src='./
offlightbulb.png'">Turn the light off</button>:

Function: Same as above, but switches the image back to off.

Explanation: Clicking this button changes the src back to offlightbulb.png.

***Note:* Please try these codes by yourself many times to get used to it**

How can I change HTML style by JavaScript?

```
document.getElementById("demo").style.fontSize = "۳۵px";
```

document Refers to your whole HTML page.

- .getElementById("demo") → Finds the element in your page with the ID "demo".
- Example: Hello
- .style → Lets you change the CSS style of that element.
- .fontSize → Targets the font size of the text inside that element.
- = "۳۵px"; → Sets the font size to **۳۵ pixels**.

How can I hide HTML elements?

```
document.getElementById("demo").style.display = "none";
```

.style.display : Controls how the element is displayed (its CSS display property).

- = "none"; : Hides the element completely (it disappears from the page layout).

How can I show HTML elements?

```
document.getElementById("demo").style.display = "block";
```

display = "block" : makes the element visible again (as a block element, like a <p> or).

The <script> Tag

In HTML, JavaScript code is inserted between <**script**> and </**script**> tags.

First what's the structure of a website?

The structure of the website should look like this :

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Website</title>
</head>
<body>

  <h1>Hello, World!</h1>
  <p>This is my first simple website.</p>

</body>
</html>
```

so the website should look like this :

Hello, World!

This is my first simple website.

<!DOCTYPE html> → tells the browser this is HTML.

- **<html>** → starts the HTML document.
- **<head>** → hidden info (title, styles, scripts).
- **<title>** → the text you see on the browser tab.
- **<body>** → everything the user sees on the page.
- **<h\>** → big heading.
- **<p>** → a paragraph of text.

The arrangement should look like this:

```
<!DOCTYPE html>
<html>
<head>

</head>
<body>

  <h1>Hello, World!</h1>
  <p>This is my first simple website.</p>

</body>
</html>
```

Where do I add the <script></script> tags?

A website has ٢ main parts:

<head> → info (title, styles, meta)
<body> → what people see

📌 Where to put <script>?

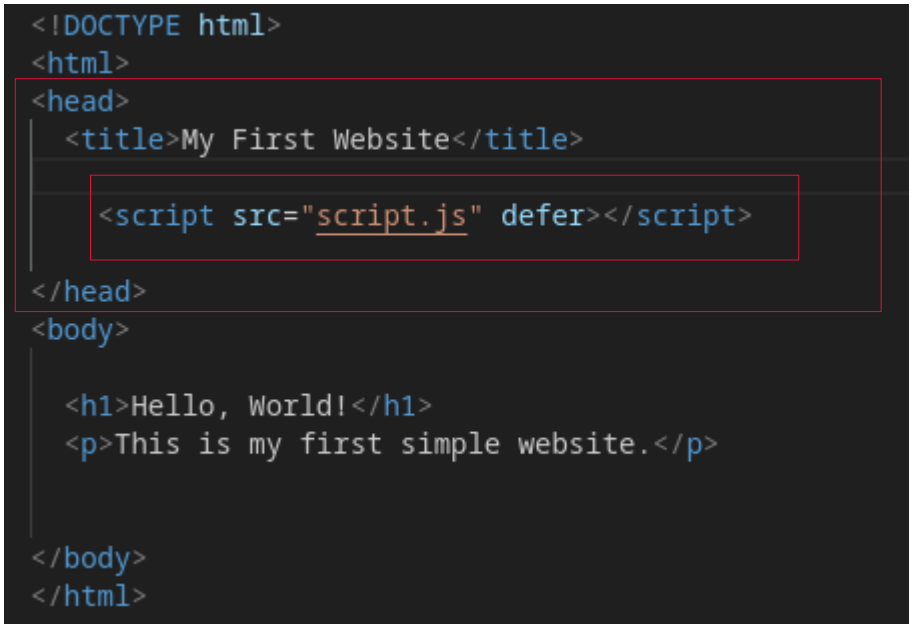
Put it at the **end of <body> (Most common method)**

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Website</title>
</head>
<body>

  <h1>Hello, World!</h1>
  <p>This is my first simple website.</p>

  <script>
    document.getElementById("demo").innerHTML = "My First JavaScript";
    </script>

</body>
</html>
```

Other way (advanced):

Put in <**head**> but add defer

☞ so it waits until page loads.



```html
<!DOCTYPE html>
<html>
<head>
  <title>My First Website</title>

    <script src="script.js" defer></script>

</head>
<body>

  <h1>Hello, World!</h1>
  <p>This is my first simple website.</p>


</body>
</html>
```

Scripts can be placed in the <**body**>, or in the <**head**> section of an HTML page, or in both.

Small definition:

What is the difference between <head> & <body> ?

<**head**> → hidden info, not shown.

Title (tab name), meta, CSS/JS links.

<**body**> → visible content.Text, images, buttons, links.

How to Use External JavaScript Files?

```
<script src="myScript.js"></script>
```

this codes adds an external JavaScript file
lets see how it works:
First step we have to create the folder:

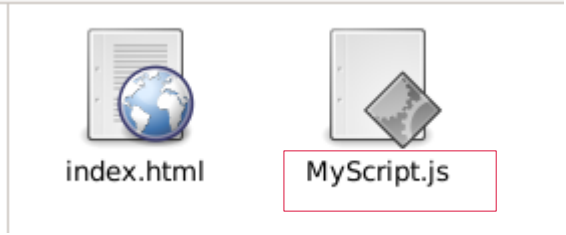
Second inside the folder, create a file called index.html.:

Third Add this code inside the index file:

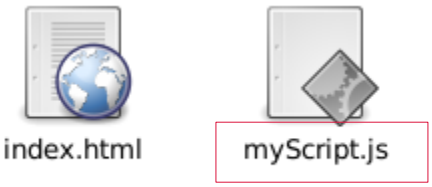
```
File  Edit  Search  View  Document  Help
<!DOCTYPE html>
<html>
<head>
  <title>External JS Example</title>
</head>
<body>

<h2>Demo: External JavaScript</h2>

<p id="demo">This is the old text.</p>
<button onclick="myFunction()">Click Me</button>

<!-- Link to external JS file -->
<script src="myScript.js"></script>

</body>
</html>
```

Fourth in the same folder, create another file called myScript.js
Make sure the “m” is small and the “S” is capital because when you add it to visual code it might be case sensitive

it shouldn’t look like this



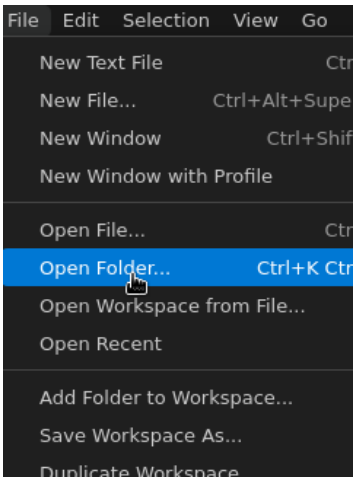
it should look like this



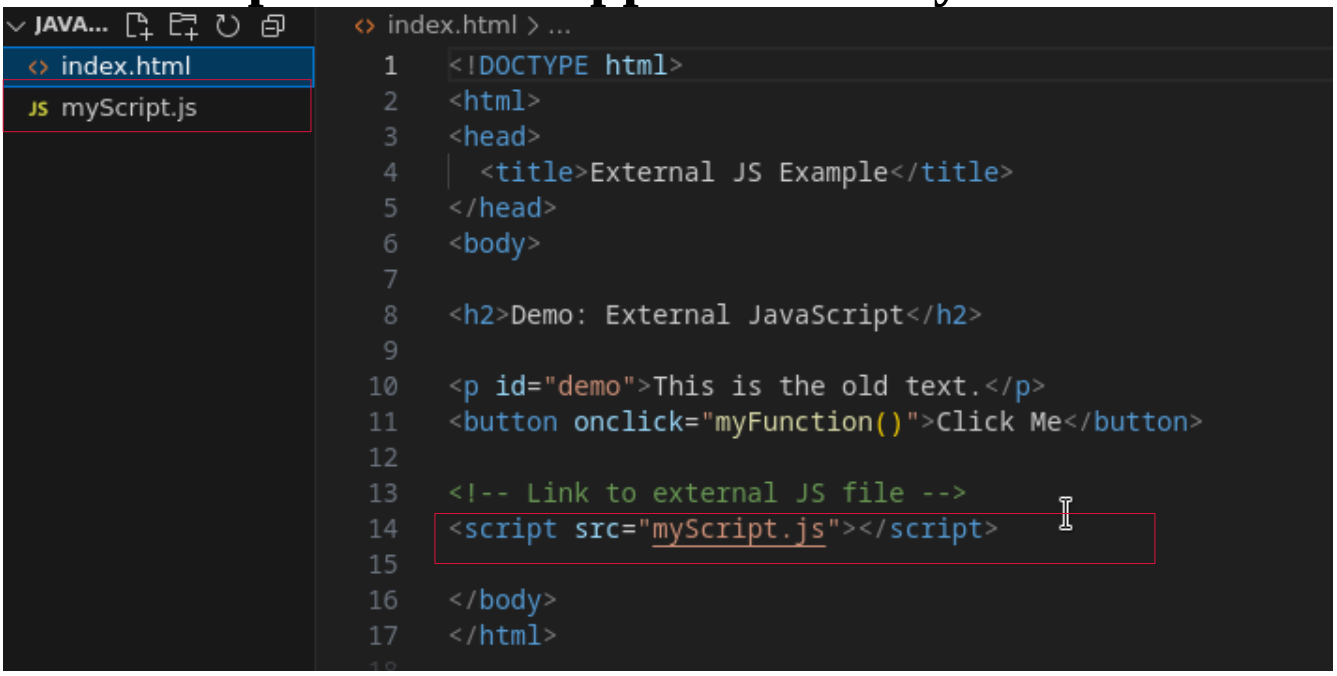
add this code to the second file:

```
File  Edit  Search  View  Document  Help
function myFunction() {
  document.getElementById("demo").innerHTML = "This text is changed by
}
```

Step five open visual code
Open the whole folder (JavaScript tutorial) in VS Code (not just one file).

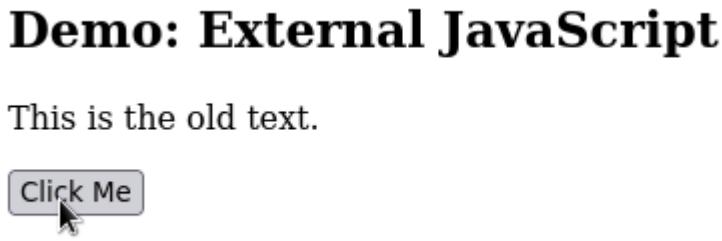


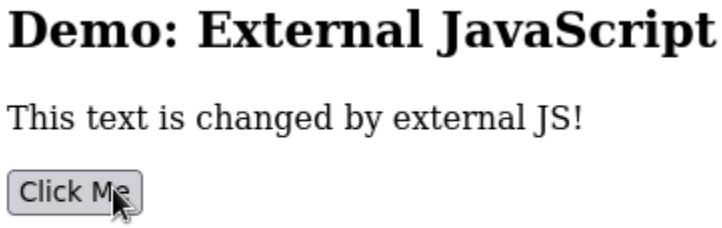
Let me explain what happened exactly



As you can see we linked “myScript.js” file with the other index file

now you can easily open with live server and you could see the result





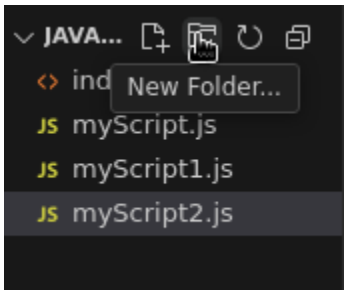
Advantages of External JavaScript

١. **Separation** → HTML = structure, JS = behavior (less mess).

٢. **Clarity** → Easier to read and fix (HTML isn't mixed with JS).

٣. **Speed** → Browser can save (cache) the JS file → faster loading next time.

You can also use several script tags as you want
And you can add new files and folders as you want using visual code



External References

You can add a website URL to your code like this:

```
<script src="https://www.Esther'sWebsite.com/js/myScript.js"></script>
```

You can add a file path it should look like this:

```
<script src="/js/myScript.js"></script>
```

If the file has no path(beside the html) it should look like this:

```
<script src="myScript.js"></script>
```

JavaScript provides several ways to display data

١. Writing into an HTML element (e.g., innerHTML, innerText**)**

innerHTML inserts HTML content.

innerText inserts plain text (ignores HTML tags).

The code is : document.getElementById(“id”)

٢. Writing into the HTML output (using document.write()**)**

Note : If used after the document is loaded, it can overwrite the entire page.

٣. Writing into an alert box (window.alert()**)**

Used to display a message in a pop-up dialog box.

٤. Writing into the browser console (console.log()**)**

console.log() displays the message to the console

its useful for debugging

Using innerHTML

document.getElementById(“id”) Then use the innerHTML property to change the HTML content of the HTML element:

Note: Changing the innerHTML property of an HTML element is the most common way to display data in HTML.

Using innerText :

document.getElementById(“id”).innerText

What is the difference between innerText and innetHTML?

- innerHTML

Returns or sets the HTML code inside an element.

It includes tags (like <b>, <i>, <p>, etc.).

If you set innerHTML, the browser will interpret it as HTML and render it.

-innerText

Returns or sets only the visible text of an element.

It ignores tags and doesn’t render them.

It considers CSS styling (like display:none, hidden text won’t show).

Lest see an example:

if we used innerHTML :

```
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML = "<b>Hello</b> World";
</script>
```

Hello World

if we used innetText :

```
<p id="demo"></p>

<script>
document.getElementById("demo").innerText = "<b>Hello</b> World";
</script>
```

Hello World

Using document.write()

```
<script>
document.write(٥ + ٦);
</script>
```

Note : Using document.write() after an HTML document is loaded, will **delete all existing HTML**

Lets see an example:

This is a normal page :

```
<!DOCTYPE html>
<html>
<body>

<h1 id="title">My Page</h1>
<p id="text">This is my content.</p>

</body>
</html>
```

This is the page without using **document.write()**:

```
<!DOCTYPE html>
<html>
<body>

<h1 id="title">My Page</h1>
<p id="text">This is my content.</p>

<button onclick="document.getElementById('text').innerHTML = 'Text
Changed ✅';">
Click Me
</button>

</body>
</html>
```

This is while using **document.write()**:

```
<!DOCTYPE html>
<html>
<body>

<h1>My Page</h1>
<p>This is my content.</p>

<button onclick="document.write('Ops 😅 everything is gone!')">Click
Me</button>

</body>
</html>
```

This is after clicking :

Note : The document.write() method should only be used for testing.

Using window.alert():

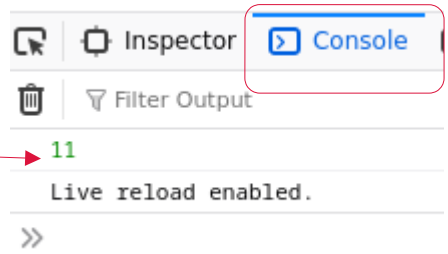
You can use an alert box to display data:

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Web Page</h1>
<p>My first paragraph.</p>

<script>
window.alert(5 + 6);
</script>

</body>
</html>
```

Note: you can skip window word and just use alert()

Using console.log():

```
<!DOCTYPE html>
<html>
<body>

<script>
console.log(5 + 6);
</script>

</body>
</html>
```



Note:For debugging purposes, you can call the console.log() method in the browser to display data.

Using JavaScript Print:

The only way to print a web page in JavaScript is with **window.print().**

Example: **<button onclick="window.print()">Print this page</button>**

```
/* These rules apply only when printing */
@media print {

  /* Hide all buttons when printing */
  button {
    display: none;
  }

  /* Hide all elements with class="no-print" */
  .no-print {
    display: none;
  }

  /* Force all text to be black in print */
  body {
    color: black;
  }
```

you can use CSS print rules to hide or show elements when printing:

@media print { ... } → means: “use these styles only when the page is being printed.”
button { display: none; } → hides all buttons in the printed page.
no-print { display: none; } → hides any element with the class no-print.
body { color: black; } → makes all text print in black, even if it’s red, blue, etc. on the screen.

Syntax rules

in JavaScript,(var, let, and const) are all used to declare variables (to store data/values).

Lets start with var :

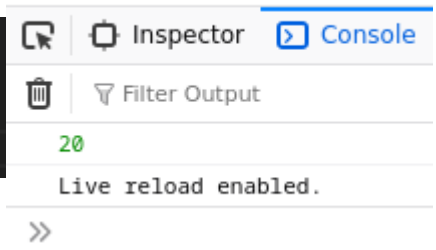
Function-scoped (it belongs to the whole function, not just a block).

Can be **redeclared** and **reassigned**.

Hoisted (moved to the top of its scope automatically).

I.e :

```
<script>var a = 10;
 var a = 20; // ✔ allowed (redeclared)
 console.log(a); // 20
 </script>
```



Second let:

Block-scoped (exists only inside { ... }).

Can be **reassigned**, but **not redeclared** in the same scope.

Not hoisted in the same way as var.

I.e:

```
<script>let b = 10;
 b = 20; // ✔ allowed
 let b = 30; // × Error (same scope redeclare)
 </script>
```

Third const:

Block-scoped.
Cannot be **reassigned** or redeclared.
Good for values that never change.

I.e:

```
 <script>
 const c = 10;
c = 20; // × Error
  </script>
```

JavaScript Values

JavaScript has two types of values:

Literals → fixed values

Variables → values that can change

JavaScript Literals

Numbers can be written with or without decimals.

Variables
named values that can change (like x = ١٠).**Strings** are text, written within double or single quotes:
"Hello"

'Hello'

JavaScript Variables

Variables are **containers** for **storing data values**.

Variables must be **identified** with **unique names**.

```
// Define x as a variable
let x;

// Assign the value ٦ to x
x = ٦;
```

JavaScript Identifiers

Identifiers are used to **name variables,** keywords, and functions.

The rules for legal names are the same in most programming languages:

A name must begin with

A name must begin with	A name can contain
A letter (A-Z or a-z)	A letter (A-Z or a-z)
	A number (٠-٩)
A dollar sign ($)	A dollar sign ($)
An underscore (_)	An underscore (_)

Note : Numbers are not allowed as the first character in names.

This way JavaScript can easily distinguish identifiers from numbers.

JavaScript Keywords

JavaScript enables interaction, and its keywords are reserved words that define actions in the language.

JS Keywords

(Let & const) --> make variables

ex:

```
let x = ٥;    const name="Java";
```

JS Operators

(=) --> assign value

ex:

let sum=x+y;

Math operations : (+, -, *, /,)

+ for summation

- for subtractions

* for multiplication

/ for division

JS Expressions

combo of vals, vars, ops → result

ex: (٥+٦)*١٠ → ١١٠

ex: x*١٠

ex: "java"+" "+"script" → "java script"

“” is considered as a space

By the way java script knows if + for names its joining if numbers its addition

JS Comments

// or /*...*/ --------> ignored by JS

ex:

let x=5; // note

Case Sensitive

lastName ≠ lastname

let, not Let or LET

Naming Styles

Hyphen: first-name (not allowed)

Underscore: first_name (allowed)

PascalCase: FirstName (allowed) also called (**Upper Camel Case)**

camelCase (common): firstName(allowed) also called (**Lower Camel Case)**

Note: Hyphens are not allowed in JavaScript. They are reserved for subtractions.

JavaScript Statements

A statement = \ instruction.

Ex:

```
let x, y, z; // declare
x = 0;       // assign
y = 7;       // assign
z = x + y;   // calc
```

S Programs

Program = list of statements.

Run top to bottom (order matters).

In HTML, JavaScript programs are executed by the web browser.

Statement Parts = list of instructions

Built from → values, operators, expressions, keywords, comments.

Ex:

```
document.getElementById("demo").innerHTML = "Java script";
```

Semicolons ;

End statements (optional but recommended).

Can write many on one line:

```
a=0;
b=7;
c=a+b;
```

or

```
a=0; b=7; c=a+b;
```

Note :JavaScript programs (and JavaScript statements) are often called JavaScript code.

White Space

Extra spaces ignored.

Good style → spaces around operators:

Ex :

let x = y + z;

let x=y+z; ← spaces are ignored here

Line Breaks

Break long lines after operators:

```
document.getElementById("demo").innerHTML =
"Java script!";
```

Note: JavaScript ignores multiple spaces. You can add white space to your script to make it more readable.

Code Blocks {}

Group statements with { }.

Used in functions, loops, if, etc.

Note : The purpose of code blocks is to define statements to be executed together.

Ex:

```
function myFunction()
{ document.getElementById("demo١").innerHTML = "Hello";
 document.getElementById("demo٢").innerHTML = "How are you?"; }
```

JavaScript Keywords

(Reserved Words)

Reserved → can’t be used as variable names.

Special words → define actions.

١. var → old variable

Declares variable (function-scoped, old style).

Example:

```
Var name = “Java”;
```

٢. let → block variable

Declares variable (block-scoped, modern).

Can change value.

```
let age = ٢٠;

age = ٢١; // ok
```

٣. const → block constant

Declares constant (block-scoped).

Value cannot change.

```
const PI = ٣.١٤;
// PI = ٣.١٤١٥٩ ✕ error
```

٤. if → condition

Runs code if condition is true.

```
if (age >= ١٨) { console.log("Adult"); }
```

٥. switch → **cases**

Many conditions → choose one.

```
switch(color) {
  case "red": console.log("Stop"); break;
  case "green": console.log("Go"); break;
  default: console.log("Unknown");
}
```

٦. for → loop

Repeats code set times.

```
for (let i=٠; i<٣; i++) {

console.log(i);

}
```

٧. function → declare function

Groups code to reuse later.

```
function greet() {
  console.log("Hello");
}
greet(); // call function
```

٨. return → exit function

Sends a value back.

```
function add(a, b) {
  return a + b;
}
console.log(add(٢,٣)); // ٥
```

٩. try → handle errors

Catches errors, avoids crash.

```
try {
  console.log(x); // ✕ error
} catch (err) {
  console.log("Error happened!");
}
```

JavaScript comments

What are comments?

Notes inside code.

Ignored by JS (not executed).

Used to explain, make readable, or disable code during testing.

Types of Comments

١. Single-line (//)

Starts with //.

Everything after // until the end of line = ignored.

Example:

```
<script>
// Change heading
document.getElementById("myH").innerHTML = "My First Page";

// Declare x and give it 5
let x = 5;   // inline comment (after code)
</script>
```

٢. Multi-line (/* ... */)

Starts with /* and ends with */.
Everything in between = ignored.
Can span multiple lines.

Example:

```
<script>
/*
This block changes:
1. Heading text
2. Paragraph text
*/
document.getElementById("myH").innerHTML = "My First Page";
document.getElementById("myP").innerHTML = "My first paragraph.";
</script>
```

(Variables & Operators)

Variables = containers for data

Declared in ٤ ways

Modern (preferred): let, const

Old: var, or auto (not recommended).

let vs const vs var

const → value can’t change.

let → value can change.

var → old, avoid using.

auto (x=٥) → bad practice.

Data Types (٨ main)

String → "Hello"

Number → ١٠, ٧.٥

Big Integer → large numbers

Boolean → true/false

Undefined → declared, no value

Null → “nothing”

Symbol → unique identifier

Object → key-value pairs

Also: Array ([]), Date (new Date()).

An **array** is a special type of object that stores **lists of values** (in order).

A **Date** object represents a point in time (date + time).

Identifiers (variable names)

Can use letters, digits, _, $.

Must start with letter / _ / $.

Case-sensitive (car ≠ Car).

Cannot use JS keywords (let, if, etc.).

Assignment

= means assign (not equal).

Example: x = x + 0 → adds 0 to x.

For equality checks: use == or ===.

== (loose equality)

Means: *“Are they similar?”*
Even if they wear different clothes.

Example:

0 (number)

"0" (string / text)

→ true they’re similar

=== (strict equality)

Means: *“Are they exactly the same?”*
Same thing, same clothes.

Example:

0 (number)

"0" (string)
→ false they’re not the same

In declaring

let car; → undefined.

let car = "Volvo"; → with value.

Also you can use many in one line:

let name="John", age=٢٠, car="BMW";

Re-declaring

var → can redeclare.

let / const → cannot redeclare.

So what does redeclare mean?

So in let you can reassign the value but you can’t redeclare it

Example:

you can say

let x = ٣

x = ٤ ← value reassigned

but you cant say

let x = ٣

let x = ٤ ← value can’t be reassigned

we can do that only using var (not recommended).

Var x = ٥

var x = ٣

Arithmetic

Works like algebra: let z = x + y;.

Strings + numbers → concatenate:

"٥" + ٢ + ٣ → "٥٢٣"

٢ + ٣ + "٥" → "٥٥"

Special Characters

$ → valid, often used in jQuery.

_ → valid, often used for private vars.

→ valid, used for truly private vars

Golden Rules

Always declare variables.

Use const by default.

Use let if value will change.

Avoid var.

JavaScript let

Rules of let:

١- **Block scope** → only works inside { }.

```
<script>
{
  let x = 2;
}
console.log(x); // × error
</script>
```

NOT CORRECT

```
<script>
{
  let x = 2;
console.log(x);
}
</script>
```

CORRECT

But we can use this

for var

٢- Must be declared before use

```
<script>
console.log(x); // × error
let x = 5;
</script>
```

NOT CORRECT

```
<script>
let x = 5;
console.log(x);
</script>
```

CORRECT

٣- Cannot be redeclared in the same block

```
<script>
let x = 1;
let x = 2;  // × error
</script>
```

NOT CORRECT

```
<script>
let x = 1;
{
  let x = 2;  // ✔ new block, new x
}
</script>
```

CORRECT

٤- **Can be reassigned** (value can change).

```
<script>
let car = "Volvo";
car = "BMW"; // ✔ works
</script>
```

What is global scope?